

Executive Summary

Bosta is moving from old nightly data transfers to a modern, near-real-time streaming system for moving data from MongoDB to Redshift.

1. The Problem

The current system updates data only once every 24 hours by running heavy nightly scripts. This causes three issues:

- **Stale Data:** Reports are up to a day old.
- **System Stress:** Nightly transfers slow down the live production database.
- **Fragile Tech:** Maintaining two different types of scripts (full and partial updates) is messy and breaks easily.

2. The Solution

We are adopting **CDC (Change Data Capture)** using **Debezium and Kafka**.

- **How it works:** Instead of copying whole tables at night, the new system automatically copies individual data changes within seconds of them happening.
- **Why this tool?**

Alternatives like AWS DMS and Estuary Flow were rejected because Kafka is more mature, flexible, and handles sudden data changes better.

AWS DMS (Database Migration Service)

- weak schema evolution handling (it depends on how frequent the schema evolution happens, I still see this is a valid option if this point is not frequent)
- AWS vendor lock-in risk.

Estuary Flow

- A modern managed CDC platform that bundles source capture, buffering, and destination loading into one service. Promising for smaller teams, but relatively immature for large-scale production deployments, and introduces vendor dependency.

3. Benefits & Trade-offs

The Good

- **Fresh Data:** Redshift updates in seconds, not hours.
- **Safe for Production:** Debezium reads exclusively from MongoDB's oplog — never querying collections directly. It reads from a backup database copy (**Secondary Replica**), meaning zero slowdowns for live users.
- **Flexible:** If fields change or get added to the database, the system adapts without breaking.

The Risks & Fixes

- **Complexity:** Managing Kafka is harder.
Fix: Use a managed service (like AWS MSK).
- **Missed Data Risk:** If the system goes offline for too long, it might miss changes.
Fix: Keep a 72-hour history log and set up alerts for delays.
- **Slight Delay:** Data takes a few seconds/minutes to arrive.
Accepted: Fine for analytics, just not for instant transactions.

4. Monitoring & Alerts

The pipeline will be monitored 24/7. Critical alerts will be sent to the team based on severity:

Severity	Issue	Action Required
P1	Pipeline completely dead	Fix within 15 minutes
P2	The system is lagging/slow	Fix within 30 minutes
P3	Minor data quality warning	Review within 2 hours